ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 (solution development). Section A: objective questions (30) · Section B: Python and the interpreter, data types and input, selection and loops, functions, lists and dictionaries, classes and objects, and processing robot sensor data (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write all program code in Python 3 unless the question says otherwise. Indentation must be clear. Small spelling mistakes in code will not be penalised if the logic is correct.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Python programs are run by a/an …

A assembler. B linker only.
C interpreter. D slicer.

1.1.2 The function that reads text typed by the user in Python is …

A input() B print() C read() D cin

1.1.3 The result of 7 // 2 in Python is …

A 3.5 B 3 C 1 D 4

1.1.4 Which line converts the text "25" into a whole number?

A str("25") B float("25") only
C list("25") D int("25")

1.1.5 In Python, blocks of code (for example inside an if-statement) are shown by …

A indentation. B curly brackets { }.
C semicolons. D the word end.

1.1.6 range(2, 8, 2) produces the values …

A 2, 4, 6, 8 B 2, 3, 4, 5, 6, 7
C 2, 4, 6 D 8, 6, 4, 2

1.1.7 A function is defined in Python with the keyword …

A function B void C def D func

1.1.8 Which data structure stores key–value pairs such as {"left": 32, "right": 41}?

A dictionary B list
C tuple D string

1.1.9 readings = [12, 8, 15]; what is readings[-1]?

A 12 B 8 C an error D 15

1.1.10 Inside a class, the first parameter of a method that refers to the object itself is normally called …

A this B self C me D obj

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Python is compiled to bytecode and then run by the Python virtual machine.

1.2.2 Comments in Python start with //.

1.2.3 input() always returns a string.

1.2.4 elif is short for "else if".

1.2.5 A while loop can run forever if its condition never becomes False.

1.2.6 Lists in Python cannot be changed after they are created.

1.2.7 len(readings) gives the number of items in the list readings.

1.2.8 Variables created inside a function are local to that function.

1.2.9 __init__ is the constructor method of a Python class.

1.2.10 Python 2 is the version prescribed for this subject.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	IDE recommended for Python in this subject	A	inheritance
1.3.2	Data type for True or False	B	bool
1.3.3	Data type for decimal numbers	C	append()
1.3.4	Adds an item to the end of a list	D	f-string
1.3.5	Leaves a loop immediately	E	PyCharm
1.3.6	Sends a value back from a function	F	try … except
1.3.7	Handles an error without the program crashing	G	float
1.3.8	Code file that can be imported into another program	H	module
1.3.9	A class that receives attributes and methods from another class	I	break
1.3.10	Inserting values into a string, e.g. f"{temp} °C"	J	return

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PYTHON BASICS

2.1 Explain how running a Python program with an interpreter differs from compiling a C++ program. (4)

2.2 Write a Python program that asks for the distance (in metres) and the time (in seconds) a robot travelled, calculates the speed, and displays it rounded to two decimal places with the unit m/s. (5)

2.3 The user may type text instead of a number in 2.2. Explain how try … except can be used to handle this, and write the code for reading the distance safely. (4)

[13]

QUESTION 3: SELECTION AND LOOPS

3.1 A line-following robot reads a light value from 0 to 100. Write an if–elif–else statement that prints "Turn left" for values below 30, "Straight" for 30 to 70, and "Turn right" above 70. (5)

3.2 Study the code below and answer the questions.

```
count = 0
total = 0
for value in [4, 9, 2, 7, 6]:
    if value > 5:
        total += value
        count += 1
print(count, total)
```

3.2.1 What is printed? (2)

3.2.2 Rewrite the code so that it uses a while loop and an index variable instead of the for loop. (5)

3.3 Write a loop that keeps asking for a password until the user types "robot4", allowing a maximum of three tries, and prints "Access granted" or "Locked". (6)

[18]

QUESTION 4: FUNCTIONS, LISTS AND DICTIONARIES

4.1 Write a function average(values) that receives a list of numbers and returns their average. It must return 0 if the list is empty. (5)

4.2 A robot logs temperatures every hour: temps = [21.5, 23.0, 26.5, 31.0, 29.5, 24.0]

4.2.1 Write code to print the highest temperature and the hour (index) at which it happened, without using max(). (5)

4.2.2 Write code to create a new list hot containing only temperatures above 25. (3)

4.3 A dictionary stores the battery level of each robot: battery = {"R1": 80, "R2": 15, "R3": 45}

4.3.1 Write code to add robot "R4" with 100. (1)

4.3.2 Write code that prints the name of every robot with a battery level below 20. (4)

[18]

QUESTION 5: CLASSES AND A ROBOT PROGRAM

5.1 Write a class Motor with: attributes name and speed (speed starts at 0); a method set_speed(value) that only accepts values from 0 to 255 (otherwise it prints "Invalid speed"); and a method stop() that sets the speed to 0. (8)

5.2 Write code to create a Motor called "left", set its speed to 180 and then stop it. (3)

5.3 Explain TWO advantages of using classes when programming a robot with several motors and sensors. (2)

5.4 Explain the difference between a list and a dictionary, and say which one you would use to store the names of sensors together with their pin numbers. (8)

[21]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER